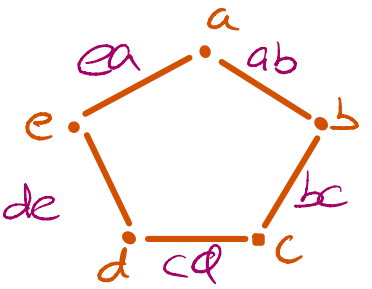
**Definition.** *A **graph** G is an ordered pair* $G = (V, E)$ *consisting of a set of **vertices** V and a set of **edges** E where*

$$E \subseteq \{\{u, v\} : u, v \in V, \text{ and } u \neq v\}.$$

size-2 subsets of V

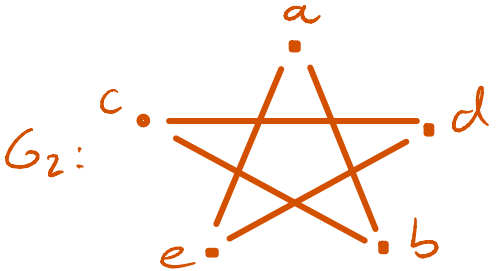
- Vertices of G is denoted by $V(G)$ and edges by $E(G)$.
- $|V|$ is called the **order** of G and $|E|$ is its **size**.
- If $u \neq v$ are vertices, we use $uv$ to abbreviate $\{u, v\}$.
- If $uv \in E$, then the vertices $u$ and $v$ are called **adjacent** or **neighbors**.
- Two edges are **adjacent** if they share some vertex.
- A vertex $v$ and an edge $e$ are **incident** if $v \in e$.
- The **null graph** is the graph with $V(G) = \varnothing$.
- Two graphs G, H are **equal** if $V(G) = V(H)$ and $E(G) = E(H)$. We denote $G = H$.

$G_1$:




$V(G_1) = \{a, b, c, d, e\}$

$E(G_1) = \{ab, bc, cd, de, ea\}$




$G_2$:

$G_1 = G_2$? Yes.

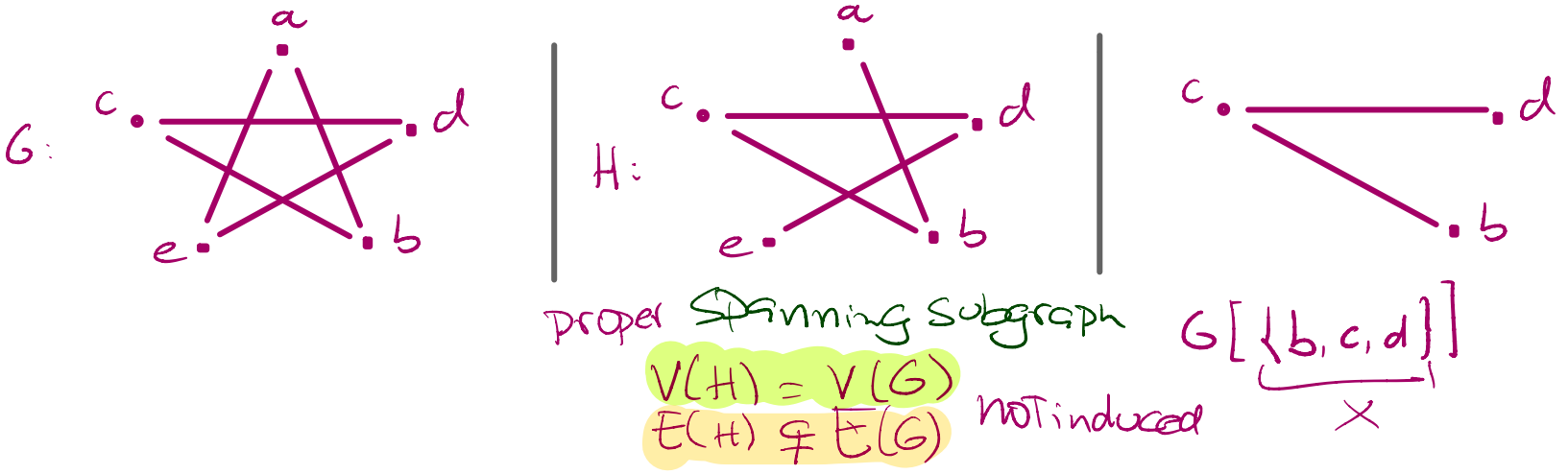
$V(G_2) = \{a, b, c, d, e\}$

$E(G_2) = \{ab, ae, bc, dc, de\}$

**Definition.** *We say that a graph* $H$ *is a* ***subgraph*** *of* $G$ *if* $V(H) \subseteq V(G)$ *and* $E(H) \subseteq E(G)$. *We denote* $H \subseteq G$.

- $H$ is a **proper subgraph** if $H \subseteq G$, but $H \neq \varnothing$ and $H \neq G$.
- $H$ is a **spanning subgraph** if $H \subseteq G$ and $V(H) = V(G)$.
- $H$ is an **induced subgraph** if $H \subseteq G$, and for all $u, v \in V(H)$, if $uv \in E(G)$ then also $uv \in E(H)$.
- If $X \subset V(G)$, then $G[X]$ denotes the induced subgraph $H$ of $G$ with $V(H) = X$.
- If $G_1, \dots G_k$ are subgraphs of $G$ such that $V(G) = V(G_1) \sqcup \cdots \sqcup V(G_k)$ and $E(G) = E(G_1) \sqcup \cdots \sqcup E(G_k)$ (disjoint unions), we say that $G$ is the **disjoint union** of the $G_i$ and denote $G = G_1 \sqcup \cdots \sqcup G_k$.

Determined by $V(H)$ and $E(G)$




proper spanning subgraph

$V(H) = V(G)$

$E(H) \subsetneq E(G)$

not induced

$G[\{b, c, d\}]$

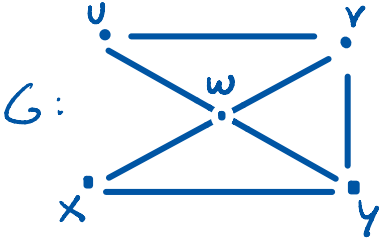
X

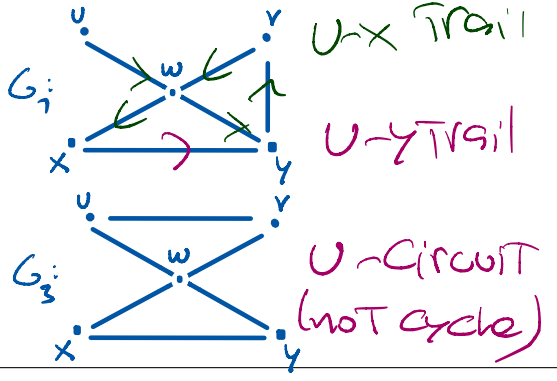
**Definition.** *A **walk** in* $G$ *is a sequence of vertices* $W = (v_0, v_1, \ldots, v_n)$ *where* $v_i v_{i+1} \in E(G)$*, for all* $0 \leqslant i < n$.

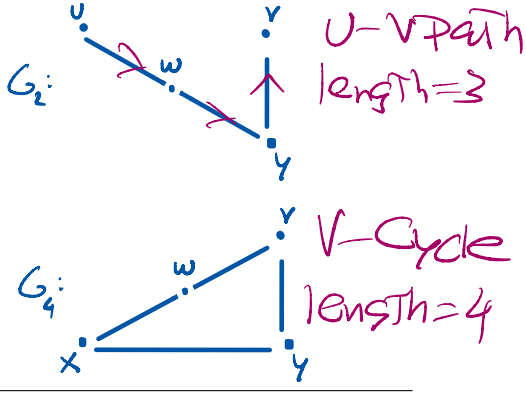
- A **trail** is a walk without repeated edges.
- A **path** is a walk without repeated vertices.
- If a {walk, trail, path} starts with $u$ and ends with $v$, it is called a $u$–$v$ {**walk, trail, path**}.
- The **length** of a walk is the number of edges.
- A $u$–$v$ {walk, trail} is **closed** if $u = v$, and it is **open** if $u \neq v$.
- A **circuit** is a closed trail of length at least 3.
- A **cycle** is a is a circuit that repeats no vertex, except for the first and last.

Path ⇒ Trail

**Example** (Identify trails, paths, circuits and cycles).



**Note:** A **drawing** of G is a sketch assigning points to V and curves to E, where endpoints of $uv$ are $u$ and $v$. These are useful yet misleading tools, hence **your proofs shouldn't include drawings, unless explicitly requested**.

**Proposition** (Theorem 1.6). *If a graph contains a $u$–$v$ walk of length $\ell$, then it contains a $u$–$v$ path of length at most $\ell$.*

Proof: Take a walk $W = (u = v_0, v_1, \dots, v_\ell = v)$

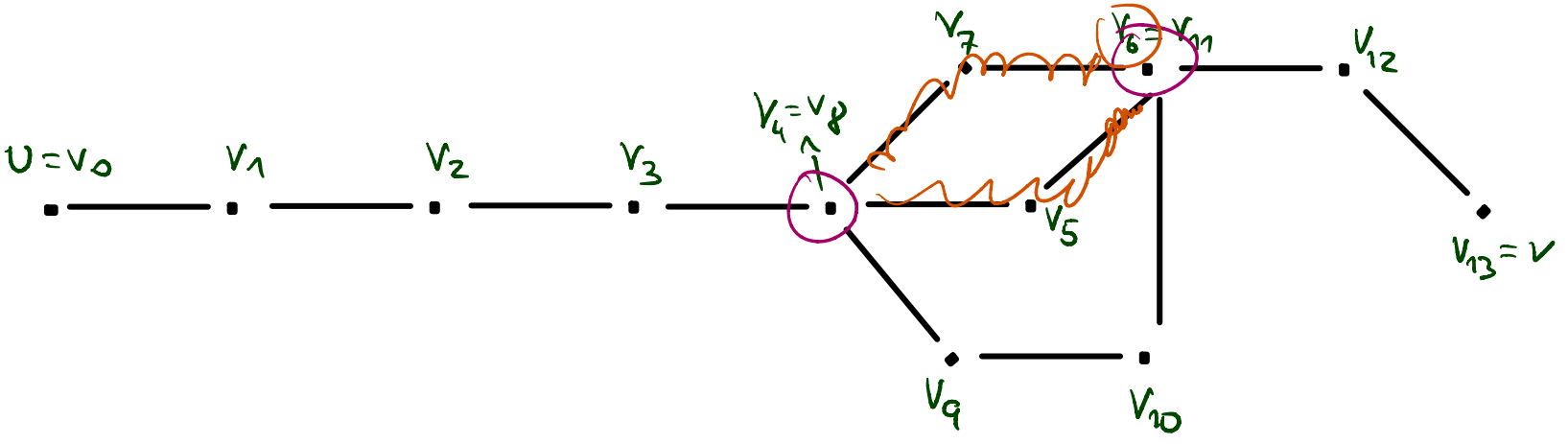
Take $i$ to be the smallest index for which $v_i = v_j$ for some $j > i$. Consider now the walk

$$W_1 = (u = v_0, v_1, \dots, v_i = v_j, v_{j+1}, \dots, v_\ell = v)$$

This is a new $u$–$v$ walk of length $< \ell$.

We continue like this until we reach a path. ∎

Alternative proof: Start with a walk $W$ of minimal length. Use the argument above to show that $W$ must be a path.


$u = v_0$
$v_1$
$v_2$
$v_3$
$v_4 = v_8$
$v_5$
$v_6 = v_{11}$
$v_7$
$v_9$
$v_{10}$
$v_{12}$
$v_{13} = v$

So far, our examples share a suspicious property: we can travel from any vertex to any other vertex.

**Definition.** *Two vertices $u, v$ in a graph $G$ are said to be **connected** if $G$ contains a $u$–$v$ path.*

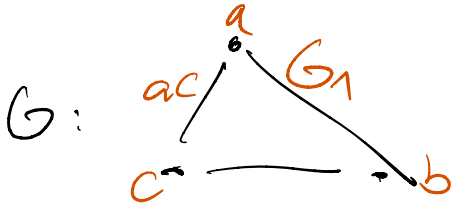
- For a vertex $u$, we allow the empty walk $W = (u)$, so that $u$ is connected to itself.
- $G$ itself is **connected** if every two vertices of $G$ are connected. Otherwise, it is **disconnected**
- A **component** of $G$ is a connected subgraph of $G$ that is not a proper subgraph of any other connected subgraph of $G$.

**Theorem.** *Each vertex and each edge of $G$ belong to exactly one component of $G$. Hence, if $G_1, \dots G_k$ are the components of $G$, we have $G = G_1 \sqcup \cdots \sqcup G_k$.*

$\longrightarrow V(G) = V(G_1) \cup \cdots \cup V(G_k)$, $E(G) = E(G_1) \cup \cdots \cup E(G_k)$

**Yoga:** If we understand the components of a graph, we understand the graph.

$G = G_1 \cup G_2$



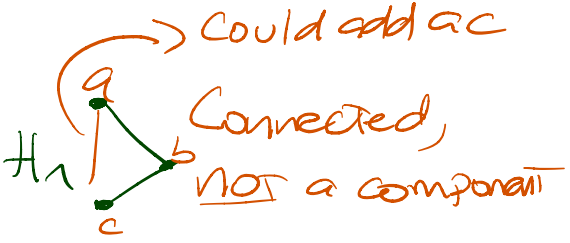

Component

$\sqcup$




Component

NOT connected



So far, our examples share a suspicious property: we can travel from any vertex to any other vertex.

**Definition.** *Two vertices $u, v$ in a graph $G$ are said to be* ***connected*** *if $G$ contains a $u$–$v$ path.*

- For a vertex $u$, we allow the empty walk $W = (u)$, so that $u$ is connected to itself.
- $G$ itself is **connected** if every two vertices of $G$ are connected. Otherwise, it is **disconnected**
- A **component** of $G$ is a connected subgraph of $G$ that is not a proper subgraph of any other connected subgraph of $G$.

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**Yoga:** If we understand the components of a graph, we understand the graph.

Proof: Equivalence relation $R$: $uRv$ iff There is a $u$–$v$ path

Exercise: $R$ is an equivalence relation.

1) $uRu$ 2) $uRv \Rightarrow vRu$

3) $uRv$, $vRw \Rightarrow uRw$

$(u, \dots, v)$ $(v, \dots, w)$ $(u, \dots v, \dots w)$ $u$–$w$ walk $\Rightarrow \exists$ path

Proposition

So we have equivalence classes of vertices.

$$V(G) = V_1 \cup \cdots \cup V_k$$

Exercise: $G[V_i]$ is a component of $G$.

So far, our examples share a suspicious property: we can travel from any vertex to any other vertex.

**Definition.** *Two vertices* $u, v$ *in a graph* $G$ *are said to be* ***connected*** *if* $G$ *contains a* $u$–$v$ *path.*

- For a vertex $u$, we allow the empty walk $W = (u)$, so that $u$ is connected to itself.
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**Theorem.** *Each vertex and each edge of* $G$ *belong to exactly one component of* $G$*. Hence, if* $G_1, \dots G_k$ *are the components of* $G$*, we have* $G = G_1 \sqcup \cdots \sqcup G_k$.

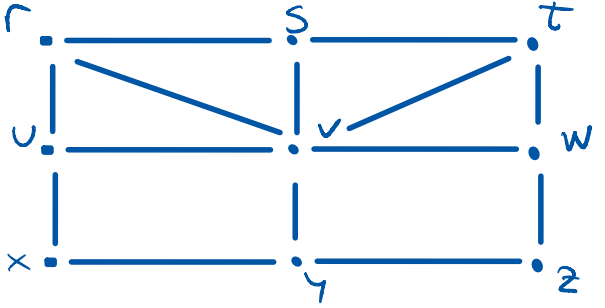
**Yoga:** If we understand the components of a graph, we understand the graph.

**Definition.** *In a connected graph* $G$*, the* ***distance*** *between vertices* $u, v$ *is*

$$d(u, v) := \text{ smallest length of any } u\text{–}v \text{ path in } G.$$

- A $u$–$v$ path of length $d(u, v)$ is called a **geodesic**.
- The **diameter** of $G$, denoted $\operatorname{diam}(G)$, is the greatest distance between any two vertices.

**Example** (Exercise 1.12)**.** *For the depicted graph* $G$*, give an example of each of the following or explain why no such example exists.*




1. *An* $x$–$y$ *walk of length* 6.
2. *A* $v$–$w$ *trail that is not a* $v$–$w$ *path.*
3. *An* $r$–$z$ *path of length* 2.
4. *An* $x$–$z$ *path of length* 3.
5. *An* $x$–$t$ *path of length* $d(x, t)$.
6. *A geodesic whose length is* $\operatorname{diam}(G)$.